DATA SPLITTING PROGRAM

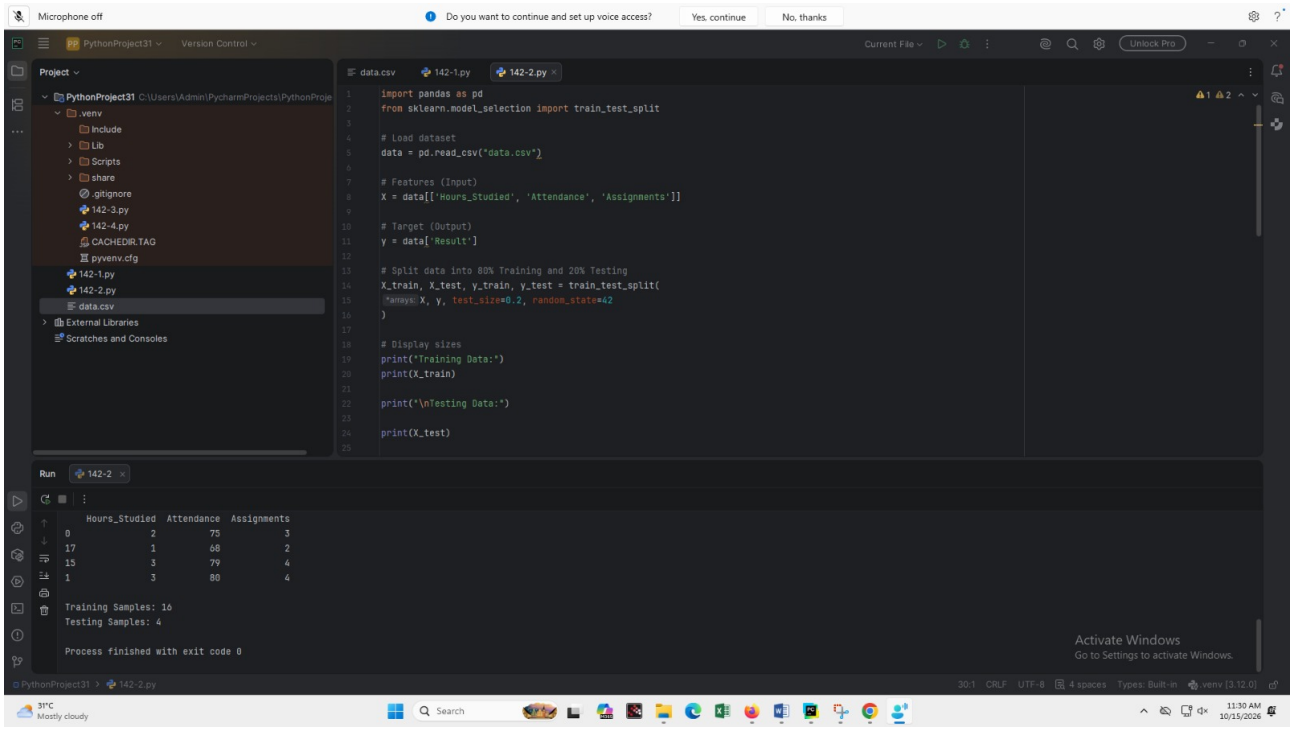
Aim

To write a program to split the given dataset into training and testing sets, so that a model can be trained on one part of the data and evaluated on unseen data.

Algorithm

1. Start the program.
2. Import the required libraries (pandas and train_test_split from sklearn.model_selection).
3. Load the dataset from the input file.
4. Separate the data into input features (X) and the target variable (y).
5. Choose the split ratio, for example 80% for training and 20% for testing.
6. Split the data into training and testing sets using train_test_split, with a fixed random_state so the split can be repeated.
7. Display the number of records in the training set and the testing set.
8. Display the training and testing data.
9. Stop the program.

Program



Data Splitting and Cross Validation Program

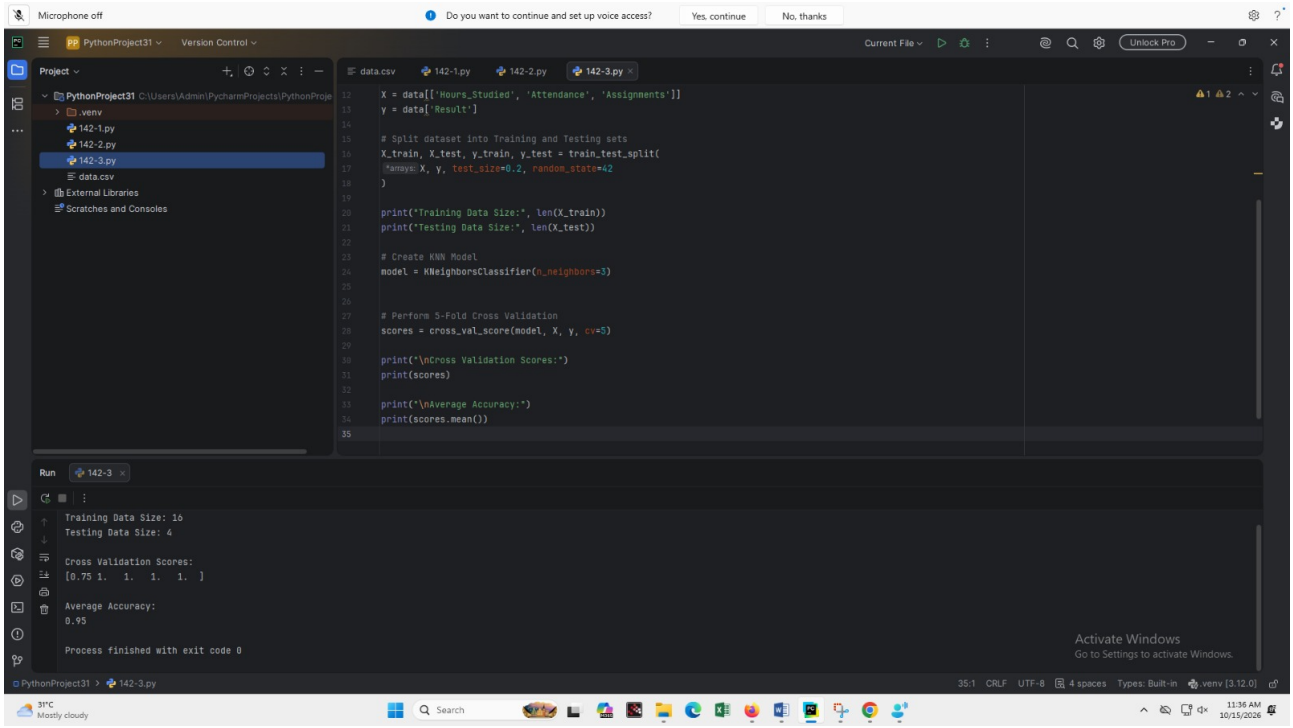
Aim

To write a program that splits the given dataset into training and testing sets and then applies k-fold cross validation, so that the model's performance is checked on different parts of the data and not on a single split.

Algorithm

1. Start the program.
2. Import the required libraries (pandas, train_test_split, KFold or cross_val_score from sklearn.model_selection, and a model such as LogisticRegression).
3. Load the dataset from the input file.
4. Separate the data into input features (X) and the target variable (y).
5. Split the data into training and testing sets using train_test_split (for example 80% training and 20% testing), with a fixed random_state.
6. Display the number of records in the training and testing sets.
7. Choose the number of folds k (for example k = 5) and create the k-fold object.
8. Divide the training data into k equal folds.
9. For each fold, use that fold as the validation set and the remaining k-1 folds as the training set.
10. Train the model on the training folds and calculate its accuracy on the validation fold.
11. Repeat steps 9 and 10 until every fold has been used once for validation.
12. Calculate the mean of the k accuracy scores to get the cross-validation score.
13. Display the score of each fold and the mean score.
14. Stop the program.

Program


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Do you want to continue and set up voice access?
Yes, continue
No, thanks
PythonProject31
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Current File
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Project
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.venv
142-1.py
142-2.py
142-3.py
data.csv
External Libraries
Scratches and Consoles
data.csv
142-1.py
142-2.py
142-3.py
X = data[['Hours_Studied', 'Attendance', 'Assignments']]
y = data['Result']
Split dataset into Training and Testing sets
X_train, X_test, y_train, y_test = train_test_split(
*arrays: X, y, test_size=0.2, random_state=42
)
print("Training Data Size:", len(X_train))
print("Testing Data Size:", len(X_test))
Create KNN Model
model = KNeighborsClassifier(n_neighbors=3)
Perform 5-Fold Cross Validation
scores = cross_val_score(model, X, y, cv=5)
print("\nCross Validation Scores:")
print(scores)
print("\nAverage Accuracy:")
print(scores.mean())
Run
142-3
Training Data Size: 16
Testing Data Size: 4
Cross Validation Scores:
[0.75 1. 1. 1. 1.]
Average Accuracy:
0.95
Process finished with exit code 0
Activate Windows
Go to Settings to activate Windows.
PythonProject31 > 142-3.py
35:1 CRLF UTF-8 4 spaces Types: Built-in .venv [3.12.0]
31°C
Mostly cloudy
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